拒绝抽样补充[1]

拒绝抽样（Rejection Sampling）[1]

目的：从$p(x)$中抽取样本[2]

方法：从$g(x)$中抽取样本，并按一定的概率接收[3]

要求：存在M，使得$p(x) \leq Mg(x)$[4]

步骤[1]

1. 从$g(x)$中抽取样本，并记为x^*

2. 计算$r = \frac{p(x^*)}{Mg(x^*)}$，并从$U[0,1]$中抽取$u$

3. 若$u \leq r$，则令$x = x^*$即为从$p(x)$中抽取的样本；否则重复1.和2.

[2]

证明：总体思路[1]

考虑x的分布函数：[2]

$$F(t) = P(x \le t) = P(x^* \le t | \text{接收}) = \frac{P(x^* \le t, \text{接收})}{P(\text{接收})}$$[3]

$$= \frac{P(x^* \le t, u \le r)}{P(u \le r)}$$

证明：分子[1]

$$P(x^* \le t, u \le r) = \int_{-\infty}^{t} \left\{ \int_{0}^{p(x^*)/[Mg(x^*)]} 1 \cdot du \right\} g(x^*) dx^*$$ [2]

$$= \int_{-\infty}^{t} \frac{p(x^*)}{Mg(x^*)} \times g(x^*) dx^* = \frac{1}{M} \int_{-\infty}^{t} p(x^*) dx^*$$

证明：分母[1]

$$P(u \le r) = P(-\infty < x^* < +\infty, u \le r)$$

$$= \int_{-\infty}^{+\infty} \left\{ \int_{0}^{p(x^*)/[Mg(x^*)]} 1 \cdot du \right\} g(x^*) dx^*$$

$$= \int_{-\infty}^{+\infty} \frac{p(x^*)}{Mg(x^*)} \times g(x^*) dx^* = \frac{1}{M} \int_{-\infty}^{+\infty} p(x^*) dx^* = \frac{1}{M}$$

[2]

证明：结论[1]

$$F(t) = P(x \le t) = \frac{P(x^* \le t, u \le r)}{P(u \le r)} = \frac{\frac{1}{M}\int_{-\infty}^{t} p(x^*)dx^*}{\frac{1}{M}}$$ [2]

$$= \int_{-\infty}^{t} p(x^*)dx^*$$

可得：x的密度函数为$p(x)$。[3]

思考[1]

条件：存在M，使得$p(x) \leq Mg(x)$在证明中有什么用？[2]